

1. What is Azure Databricks?

Azure Databricks is a cloud-based big data analytics platform provided by Microsoft Azure and built on Apache Spark. It is designed to help organizations process large amounts of data, perform data engineering, data analytics, machine learning, and build ETL (Extract, Transform, Load) pipelines efficiently.

Azure Databricks combines the power of Apache Spark with Azure's cloud infrastructure, making it easier to handle massive datasets without managing complex hardware or Spark clusters manually.

Why Was Azure Databricks Created?

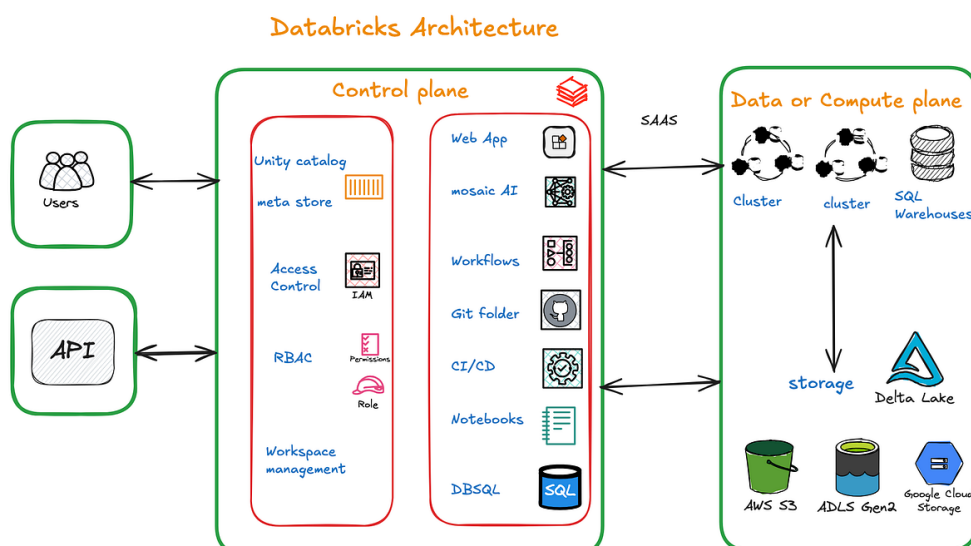
Traditional data processing systems struggle when dealing with:

- Massive datasets (TBs or PBs of data)
- Real-time data streams
- Complex data transformations
- Machine learning workloads
- Collaborative analytics projects

Azure Databricks solves these problems by providing:

- Distributed computing
- Automatic cluster management
- Collaborative notebooks
- High-speed processing
- Integration with Azure services

Azure Databricks Architecture



Azure Databricks architecture consists of:

1. Workspace

A workspace is the environment where users:

- Create notebooks
- Run queries
- Manage jobs
- Collaborate with teams

Think of it as project folder.

2. Driver Node

The Driver Node:

- Receives code from the user
- Creates Spark jobs
- Distributes tasks
- Collects results

Example:

```
df.groupBy("store_id").sum("sales")
```

The driver converts this command into tasks and sends them to worker nodes.

3. Worker Nodes

Worker Nodes:

- Process data
- Execute tasks assigned by the driver
- Return results

Example:

Worker 1 → Store A data

Worker 2 → Store B data

Worker 3 → Store C data

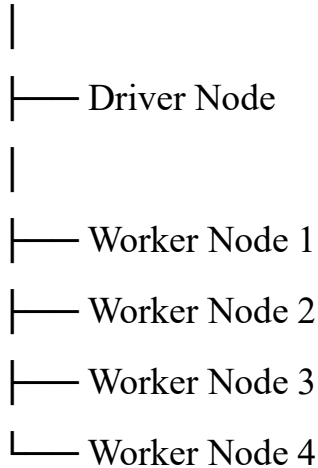
4. Cluster

A cluster is a collection of:

- One Driver Node
- Multiple Worker Nodes

Example:

Cluster



Core Components of Azure Databricks

1. Notebooks

Notebooks are interactive documents used to write and execute code.

Supported languages are Python, SQL, Scala, R.

Example:

```
df = spark.read.csv("/sales.csv", header=True)
df.show()
```

2. Clusters

Clusters provide computing power.

Types:

- **Interactive Cluster:** Used during development.
- **Job Cluster:** Created automatically when a scheduled job runs.

3. Databricks Jobs

Jobs automate tasks.

Example:

Every day at 9 AM

Read Sales Data



Clean Data



Calculate Revenue



Save Results

No manual execution required.

4. Databricks SQL

Allows SQL queries on large datasets.

Example:

```
SELECT product_name,  
       SUM(revenue)  
FROM sales  
GROUP BY product_name;
```

5. Delta Lake

Delta Lake is a storage layer that improves data reliability.

Features:

- ACID Transactions
- Data Versioning
- Time Travel

➤ Schema Enforcement

Example:

```
df.write.format("delta").save("/sales")
```

What is ETL in Azure Databricks?

ETL means:

Extract

Collect data from:

- CSV files
- Databases
- APIs

Transform

Clean and process data.

Load

Store processed data:

Azure Databricks Workflow

Example: Retail Sales Project

Step 1

Load sales data

```
sales_df = spark.read.csv("sales.csv")
```

Step 2

Clean data

```
sales_df = sales_df.dropna()
```

Step 3

Calculate revenue

```
sales_df = sales_df.withColumn(  
    "revenue",  
    sales_df.quantity * sales_df.price  
)
```

Step 4

Join product data

```
final_df = sales_df.join(product_df)
```

Step 5

Store results

```
final_df.write.format("delta").save("/sales_metrics")
```

Step 6

Create dashboard

Using:

- Power BI
- Databricks SQL

Integration with Azure Services

Azure Service	Purpose
Azure Data Lake	Store large datasets
Azure Blob Storage	File storage
Azure SQL Database	Structured data
Azure Synapse	Data warehouse
Power BI	Visualization
Azure DevOps	CI/CD pipelines
Azure Machine Learning	ML models

2. What are advantages of using Azure Databricks?

Azure Databricks offers numerous advantages for organizations working with big data, data engineering, analytics, and machine learning. It combines the power of Apache Spark with the scalability and security of Microsoft Azure, making data processing faster and more efficient.

1. High-Speed Data Processing

Azure Databricks is built on Apache Spark, which uses distributed computing to process large datasets across multiple machines simultaneously. This significantly reduces processing time and enables organizations to analyze massive amounts of data quickly.

2. Easy Scalability

Azure Databricks can automatically scale computing resources up or down based on workload requirements. As data volume increases, additional resources can be added automatically, ensuring consistent performance without manual intervention.

3. Simplified Cluster Management

Users do not need to manually configure or manage Spark clusters. Azure Databricks automatically handles cluster creation, optimization, monitoring, and maintenance, reducing administrative effort and improving productivity.

4. Collaborative Work Environment

Azure Databricks provides interactive notebooks where multiple users can work together in real time. Data engineers, analysts, and data scientists can share code, results, and visualizations within the same workspace, improving teamwork and collaboration.

5. Seamless Azure Integration

Azure Databricks integrates easily with various Azure services, including Azure Data Lake Storage, Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics, Azure Machine Learning, and Power BI. This allows organizations to build complete end-to-end data solutions within the Azure ecosystem.

6. Support for Multiple Programming Languages

Users can write code using Python, SQL, Scala, or R within the same

environment. This flexibility allows teams with different technical backgrounds to work efficiently using their preferred programming language.

7. Efficient ETL Processing

Azure Databricks is widely used for ETL (Extract, Transform, Load) operations. It can extract data from multiple sources, transform and clean the data, and load it into data warehouses or storage systems quickly and efficiently.

8. Advanced Analytics and Machine Learning

The platform supports machine learning libraries such as TensorFlow, Scikit-learn, PyTorch, and MLflow. Users can build, train, track, and deploy machine learning models directly within Azure Databricks.

9. Delta Lake Support

Azure Databricks includes Delta Lake, which provides features such as ACID transactions, schema enforcement, data versioning, and time travel capabilities. These features improve data reliability, consistency, and governance.

10. Real-Time Data Processing

Azure Databricks supports both batch processing and stream processing. Organizations can analyze real-time data from sources such as IoT devices, Event Hubs, Kafka, and application logs, enabling faster business decisions.

11. Enhanced Security

Azure Databricks provides enterprise-grade security features, including Azure Active Directory integration, role-based access control, encryption at rest and in transit, and network security controls to protect sensitive data.

12. Cost Optimization

Features such as auto-scaling and auto-termination help reduce cloud infrastructure costs. Resources are used only when needed, preventing unnecessary spending.

13. High Availability and Reliability

Since Azure Databricks runs on Microsoft's cloud infrastructure, it offers high availability, fault tolerance, and automatic recovery mechanisms, ensuring minimal downtime and reliable performance.

3. What is a Workspace?

A Workspace in Azure Databricks is the central environment where users create, organize, manage, and collaborate on data engineering, data analytics, and machine learning projects. It serves as the main interface through which users interact with Azure Databricks resources such as notebooks, clusters, jobs, libraries, dashboards, and data files.

Think of a workspace as a digital project environment where all team members can access and work on shared data processing and analytics tasks. It provides a secure and collaborative platform that helps users develop, test, execute, and monitor data workflows in one place.

Components of a Workspace

1. Notebooks

Notebooks are one of the most important components of a workspace.

A notebook is an interactive document where users can:

- Write code
- Execute commands
- Visualize results
- Document their work

Azure Databricks notebooks support:

- Python
- SQL
- Scala
- R

2. Clusters

A cluster is a group of virtual machines used to process data.

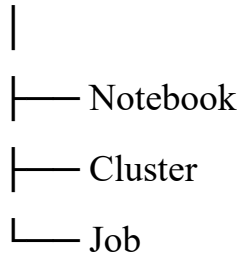
Within the workspace, users can:

- Create clusters
- Start clusters
- Stop clusters
- Monitor cluster performance

Clusters provide the computing power required to execute notebooks and process large datasets.

Example:

Workspace



When a notebook runs, it uses the resources available in a cluster.

3. Jobs

Jobs are automated tasks that execute notebooks or scripts.

Users can schedule jobs to run:

- Daily
- Weekly
- Monthly
- At specific times

Example:

Every Day at 8 AM



Load Sales Data



Transform Data



Generate Report

The workspace stores and manages these scheduled jobs.

4. Libraries

Libraries contain external packages and dependencies used by notebooks.

5. Repositories (Repos)

Repos allow integration with Git platforms such as:

- GitHub
- Azure DevOps
- GitLab
- Bitbucket

This enables:

- Version control
- Team collaboration
- Code backup
- Change tracking

Developers can manage project code directly from the workspace.

6. Dashboards and Visualizations

The workspace allows users to create visual reports and dashboards.

Examples:

- Revenue reports
- Attendance dashboards
- Product performance charts
- KPI reports

These visualizations help organizations make data-driven decisions.

7. Data Management

The workspace provides access to various data sources, including:

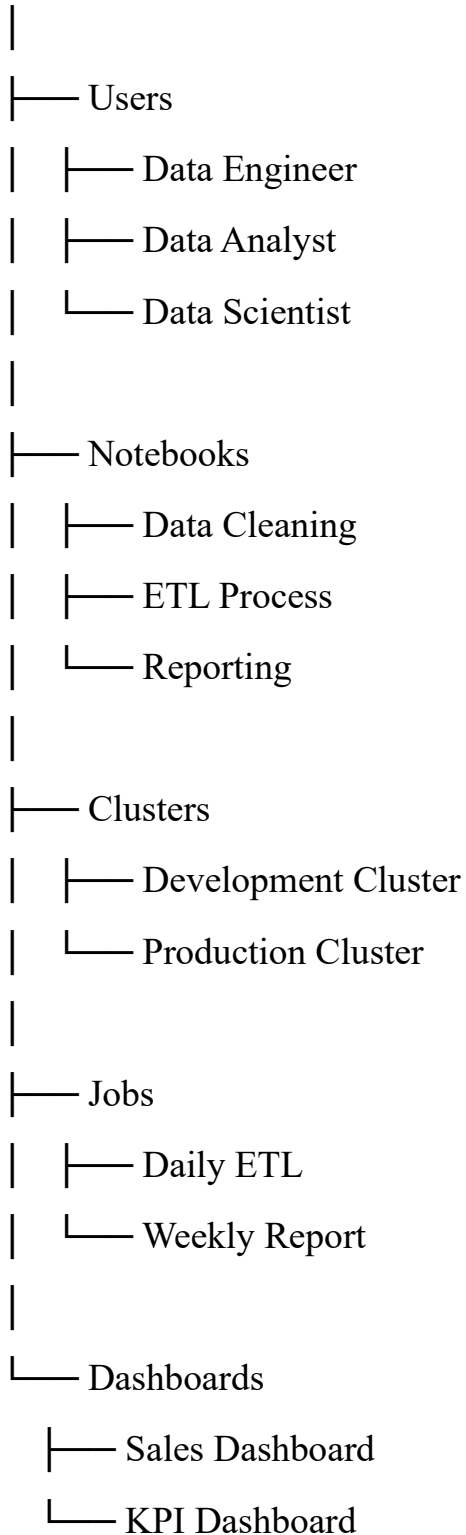
- Azure Data Lake Storage
- Azure Blob Storage
- Azure SQL Database
- Delta Tables
- CSV files
- JSON files

Users can read, transform, and write data from within the workspace.

Workspace Structure

A typical Azure Databricks workspace may look like:

Workspace



4. What is EDA (Exploratory Data Analysis)?

Exploratory Data Analysis (EDA) is the process of examining, understanding, and summarizing a dataset before performing data modeling, machine learning, or advanced analytics. It is one of the most important steps in the data analysis lifecycle because it helps analysts understand the structure, quality, and characteristics of the data.

The primary goal of EDA is to discover patterns, identify anomalies, detect missing values, understand relationships between variables, and gain insights that can guide further analysis. Through EDA, analysts can determine whether the data is clean, reliable, and suitable for business intelligence, reporting, or predictive modeling.

In simple terms, EDA is the process of exploring data to understand what it contains and how it behaves before using it for decision-making or machine learning.

EDA helps to:

- Understand the dataset structure.
- Identify missing values.
- Detect duplicate records.
- Find unusual observations (outliers).
- Discover trends and patterns.
- Analyze relationships between variables.
- Improve data quality before analysis.
- Select appropriate features for machine learning.

Objectives of EDA

The main objectives of EDA are:

1. Understand the Data

EDA helps answer questions such as:

- How many rows and columns are present?
- What are the data types?

- What information does each column contain?

2. Identify Missing Values

Missing data can affect analysis accuracy.

3. Detect Duplicate Records

Duplicate records can distort business reports.

4. Find Outliers

Outliers are unusual values that differ significantly from other observations.

5. Discover Patterns and Trends

EDA helps uncover business insights.

6. Analyze Relationships

EDA helps determine relationships between variables.

Types of EDA

1. Univariate Analysis

Analysis of a single variable.

Example:

Analyzing only the "Salary" column.

Questions:

- What is the average salary?
- What is the maximum salary?
- What is the minimum salary?

2. Bivariate Analysis

Analysis of two variables.

Example:

Salary vs Experience

Questions:

- Does salary increase with experience?
- Is there a relationship between the two variables?

3. Multivariate Analysis

Analysis involving more than two variables.

Example:

Sales, Price, Discount, and Profit

Questions:

- How do discounts affect profits?
- Which combination leads to higher sales?

Steps in EDA

Step 1: Load the Data

```
import pandas as pd  
df = pd.read_csv("sales.csv")
```

Load data into a DataFrame.

Step 2: View Sample Records

```
df.head()
```

Displays the first few rows.

Step 3: Check Dataset Information

```
df.info()
```

Step 4: Generate Summary Statistics

```
df.describe()
```

Step 5: Check Missing Values

```
df.isnull().sum()
```

Step 6: Check Duplicate Records

```
df.duplicated().sum()
```

Step 7: Detect Outliers

```
df.boxplot(column="sales")
```

Step 8: Correlation Analysis

```
df.corr(numeric_only=True)
```

Benefits of EDA

1. Improves data quality.
2. Detects errors early.
3. Identifies missing values and outliers.
4. Helps select relevant features for machine learning.
5. Reduces the risk of inaccurate analysis.
6. Provides a deeper understanding of the dataset.

EDA Example: Employee Attendance Dataset

Employee	Work Hours	Tasks Completed
John	8	20
David	6	8
Sarah	9	25

EDA can help identify:

- Top performers
- Employees with low productivity
- Average work hours
- Attendance trends
- Frequent absentees